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Economic evaluation of a multi-component obesity prevention intervention in Chinese primary schools

Shiyu Yan[1], Shuang Zhou[1], Xiru Guo[1], Jinlang Lyu[1], Hui Wang[1], Hai Fang[2*] and Hai-Jun Wang[1*]

Abstract

Background Health economic evaluations of childhood obesity interventions are critical for determining the economic feasibility, sustainability, and scalability, thereby informing their policy relevance. This study conducted a comprehensive economic evaluation of a multi-component, App-assisted obesity prevention intervention—compared with usual care—to assess its value for widespread implementation.

Methods The Diet, Exercise, and Cardiovascular Health–Children (DECIDE-Children) intervention was a cluster randomized clinical trial involving 24 schools across three socioeconomically distinct regions in China, targeting children aged 8 to 10 years. Schools were randomly assigned (1:1) to either the intervention group or the control group (usual health education). The intervention included school, family, and student components supported by a mobile application for health education, behavior monitoring, weight management, and feedback. A societal perspective was adopted for the economic evaluation. Total costs included direct financial expenditures (e.g., materials, equipment) and labor costs. Incremental cost-effectiveness ratios (ICERs) were calculated for primary obesity-related outcome measure. Using an obesity progression model, we estimated the number of adult obesity cases prevented up to age 65, quality-adjusted life years (QALYs) gained, incremental cost-utility ratio (ICUR), cost–benefit ratio (CBR), and net benefits.

Results The total cost of implementing the 1-year intervention was 13,769.74 USD, averaging 19.53 USD per student in the intervention group. Compared with usual care, the ICERs were as follows: 42.46 USD (95% CI 29.15 to 78.12) per 1-unit reduction in BMI; 11.49 USD (95% CI 7.81 to 21.70) per 0.1-unit reduction in BMI *Z*-score; 18.60 USD (95% CI 10.67 to 69.75) per 1% reduction in body fat percentage; 11.98 USD (95% CI 6.93 to 45.42) per 1-cm reduction in waist circumference. The ICUR was 4644.42 USD per QALY gained, and the CBR was 0.84. National implementation was projected to yield 419,040 QALYs gained and USD 1.86 billion in net benefits. Sensitivity analyses supported the robustness of findings.

Conclusions The DECIDE-Children intervention is a cost-effective strategy for childhood obesity prevention, providing significant public health and economic benefits. Its integration of school, family, and digital components offers a scalable, sustainable model for addressing childhood obesity in China and other developing settings.

Trial registration ClinicalTrials.gov Identifier NCT03665857.

*Correspondence:
Hai Fang
hfang@hsc.pku.edu.cn
Hai-Jun Wang
whjun@pku.edu.cn

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Keywords Economic evaluation, Children, Obesity, Prevention, Intervention

Background

Childhood obesity is one of the most important issues in global health challenges [1, 2]. According to the World Obesity Federation, the number of obese children worldwide is projected to increase from 175 million in 2020 to 380 million by 2030 [3]. This rising trend is particularly concerning in low- and middle-income countries, including China [4]. In China, the prevalence of overweight and obesity among school-aged children increased from 15.5% in 2010 to 24.2% in 2019, and is estimated to reach 31.8% by 2030 [5, 6]. Childhood obesity is associated with an increased risk of metabolic and cardiovascular diseases, as well as musculoskeletal disorders, both in the short term and later in adulthood [7–12]. In addition to its health consequences, childhood obesity imposes a substantial economic burden on society [13]. According to a meta-analysis, the pooled estimate of increased annual total direct medical costs attributable to childhood overweight and obesity was 237.55 USD per capita (2022 international dollars) [14]. Indirect costs—including caregiver absenteeism, productivity loss, and lifelong medical expenses associated with obesity persisting into adulthood—were considerably higher than direct healthcare expenditures [14]. Notably, evidence from the USA indicates that an obese 10-year-old child who remains obese into adulthood would incur an additional $12,660–19,000 in lifetime direct medical costs compared with peers who maintain a healthy weight [15]. Therefore, early prevention of childhood obesity is not only of critical importance for improving health outcomes but also represents a highly cost-effective public health strategy that warrants urgent and widespread implementation.

With the growing burden of childhood obesity, school-based preventive interventions have attracted increasing attention as a practical and scalable solution in real-world settings. Evidence suggests that these interventions are promising in reducing overweight and obesity among school-aged children [16, 17]. A systematic review reported that multi-component interventions—those integrating physical activity and dietary improvements—are more effective than single-component interventions in preventing childhood obesity [18]. Furthermore, interventions with strong family engagement tend to yield greater reductions in childhood obesity than those with minimal or no family involvement [19]. However, school-based interventions that extend into the family environment, especially multi-component programs, typically require greater financial and logistical resources, making them more costly and complex to implement than simpler interventions. In the context of finite public health resources, policymakers and practitioners must weigh both effectiveness and cost-effectiveness when considering large-scale adoption. Health economic evaluations are therefore critical tools to inform decisions on resource allocation and prioritize interventions that provide the greatest value for money. Despite growing evidence on what works in childhood obesity prevention, relatively few studies reported economic evaluation results [20].

Building on this evidence base, the Diet, Exercise, and Cardiovascular Health–Children (DECIDE-Children) intervention was implemented as a multi-component cluster randomized controlled trial (RCT) across 24 primary schools in three socioeconomically distinct regions in China (Beijing, Changzhi, and Urumqi). The intervention comprised three components targeting children (health education, reinforcement of physical activity, and BMI monitoring with feedback) and two components targeting the children's environment (engaging schools and families to support behavioral changes). A central feature of the intervention was the use of a mobile application to enhance parental engagement through health education, BMI and behavior monitoring, and personalized feedback, thereby strengthening family involvement. The intervention demonstrated significant effects in reducing childhood obesity, with a mean between-group difference in BMI of −0.46 kg/m^2 (95% CI −0.67 to −0.25) and a markedly lower odds of obesity (OR 0.34, 95% CI 0.18 to 0.64) [21]. Theoretically, the App-assisted approach may improve scalability and reduce implementation costs compared to traditional face-to-face interventions. However, the economic value of this effective intervention remains unclear, particularly in terms of its affordability and feasibility for broader adoption. Therefore, the objective of the present study was to conduct a comprehensive economic evaluation of the DECIDE-Children intervention from a societal perspective, providing evidence for policymakers and public health stakeholders about the intervention's potential for scale-up to address the ongoing childhood obesity epidemic at the national and global levels.

Methods

This economic evaluation was conducted and reported in accordance with the Consolidated Health Economic Evaluation Reporting Standards (CHEERS 2022) guidelines [21] to ensure transparency, comparability, and completeness in reporting. A completed CHEERS checklist is provided in the Additional file 1.

Intervention

The DECIDE-Children intervention was a cluster RCT conducted among children aged 8 to 10 years in three geographically and culturally distinct regions in China—Beijing, Changzhi, and Urumqi. In each region, eight primary schools were selected, resulting in a total of 24 schools participating in the study. Within each school, one or two grade 4 classes were recruited to ensure that each school contributed approximately 50 students to the study population. Schools were randomized in a 1:1 ratio to either the intervention or the control group, stratified by district within each region. The intervention design was based on the social-ecological model [22], targeting multiple levels of influence, including the individual (student), family, and school environments. Activities for schools included implementing policies related to obesity and providing health education training for teachers. For families, the intervention included health education activities and encouraged caregivers (mainly parents) to support and supervise their child's behavioral changes. Student-focused activities consisted of health education courses, enhancing physical activity, and regular monitoring of weight and height. Importantly, these student-focused components were embedded in the existing school curriculum, and no additional class periods were created. This design ensured that the program could be carried out without placing any extra academic or time burden on either teachers or students. A mobile application titled "Eat Wisely, Move Happily" was utilized to strengthen the connections among these intervention components, facilitating information diffusion, behavior monitoring, weight management, assessment, and feedback. To standardize intervention delivery, three manuals were developed for project staff, school teachers, and caregivers. Schools assigned to the control arm continued with their standard curriculum and routine physical education activities without any additional resources or obesity-specific components, thus incurring no intervention-related costs. The intervention was implemented for one school year, from late September 2018 to June 2019. Detailed information about the intervention has been reported previously [23, 24].

Costs of intervention

To estimate the economic impact of the intervention, we conducted a cost analysis from a societal perspective. The total intervention costs included both direct financial expenditures (e.g., materials, equipment) and labor components. To clarify cost attribution by stakeholder, we classified monetary costs (materials, equipment, and app-related operational expenses) as provider-borne costs, since they were directly funded by the research program. Labor costs reflected the opportunity cost of school staff time and were categorized under the school perspective. In the subsequent economic evaluation (cost-utility and cost–benefit analyses), we additionally considered medical costs and productivity losses associated with obesity, which correspond to the healthcare system and broader societal perspectives.

Monetary costs referred to direct expenditures incurred during the implementation period. These included spending on printed educational materials, sports supplies, and incentive gifts for children, all valued based on actual purchase prices. Importantly, the monetary costs also covered ongoing expenses associated with the mobile application, such as system maintenance, routine updates, user data processing, and data storage—costs that would be necessary for real-world replication. However, the initial app development costs were not included, as they were considered one-time sunk costs not expected to recur in future scale-up efforts.

Labor costs were estimated by translating time inputs from school personnel into monetary values. Specifically, we calculated labor costs by multiplying the average salary of school staff by the actual number of hours they dedicated to intervention-related activities, such as training sessions, health education, and coordination tasks. Time data were extracted from detailed process records maintained during the intervention. Four categories of staff were included: class teachers, principals, healthcare/health education teachers, and PE teachers. To reflect regional variation in wages, labor costs were calculated separately for the three intervention sites (Beijing, Changzhi, and Urumqi). We did not assign a monetary value to the time of caregivers (mainly parents). Although their participation was a crucial part of the intervention, it was facilitated entirely through a mobile application designed for flexible engagement. This design allowed parents to participate at their convenience outside of working hours, minimizing opportunity costs related to lost productivity. Similarly, we did not value children's time, as the intervention was intentionally integrated into the existing school schedule. Health education components were delivered during classes already allocated to health or moral education, and physical activity promotion was embedded within regular breaks or physical education sessions. Consequently, no core academic instruction was displaced, and the opportunity cost of student time was effectively negligible.

To isolate the cost of implementing the intervention in real-world settings, research- and evaluation-related costs were excluded. The average cost per student was calculated by dividing the total intervention cost by the number of participants in the intervention group at baseline. No additional costs were assumed for the control

group. All costs were converted to US dollars using the average exchange rate for 2018 (1 USD = 6.6322 CNY).

Obesity-related outcomes

Baseline and follow-up measurements on anthropometric measurements were collected in September 2018 and June 2019 by the trained outcome assessors using standardized procedures and instruments. The outcomes for the economic evaluation were the primary anthropometric indicators of the trial [23], including body mass index (BMI, calculated from height and weight), BMI *Z*-score (calculated according to the child growth standards released by the World Health Organization (WHO) [25]), body fat percentage (BF%), and waist circumference (WC). This approach, detailed a priori, ensured that the economic evaluation was not subject to post hoc outcome selection.

Cost-effectiveness analyses

We conducted the cost-effectiveness analyses based on the intention-to-treat principle, involving 705 participants in the intervention group and 687 participants in the control group. The incremental cost-effectiveness ratio (ICER) was calculated for primary anthropometric outcome indicator, representing the cost associated with achieving a one-unit change in the respective indicator. The ICER was calculated using the following formula [26]:

$$ICER = \frac{Cost_{IG} - Cost_{CG}}{Effectiveness_{IG} - Effectiveness_{CG}}$$

where:

- $Cost_{IG}$ = *Cost per student in the intervention group*
- $Cost_{CG}$ = *Cost per student in the control group*
- $Effectiveness_{IG}$ = *Outcome value (e.g., change in BMI Z-score) in the intervention group*
- $Effectiveness_{CG}$ = *Outcome value (e.g., change in BMI Z-score) in the control group*

For the anthropometric indicators, the ICER represented the expenditure per student in reducing one unit of BMI, 0.1 unit of BMI *Z*-score, 1% body fat percentage, and 1-cm waist circumference.

Cost-utility analyses and cost–benefit analyses

To estimate the long-term health and economic impact of childhood obesity prevention, we used a two-stage obesity-progression model to project the number of adult obesity cases prevented by the intervention [26, 27]. This approach was necessary due to the limited availability of longitudinal data linking childhood weight status directly to adult obesity incidence. Overweight and obesity were defined using both the Chinese BMI reference standards and WHO growth references to ensure broader applicability [25, 28]. In the first stage, we estimated the probability of transition between weight status categories (normal weight, overweight, or obese) from childhood/adolescence (ages 10–14) to young adulthood (ages 21–29), based on baseline weight status. The average age of children in our study was 9.6 years (SD 0.4). We referred to existing literature using the 10–14-year age band, which closely approximates the age of our participants [29]. In the second stage, we projected the transition from young adulthood (25–29 years) to adulthood (age 40). We assumed that BMI status remains stable after age 40, based on evidence from long-term cohort studies suggesting that adult weight trajectories are relatively stable or tend to increase further [30]. We assumed that the model was based on the observed reduction in childhood overweight and obesity prevalence after the 1-year intervention period. This reduction was applied to the initial weight distribution at baseline (ages 10–14), and the subsequent transitions to adulthood were projected using established progression probabilities [27, 29]. No additional long-term or sustained effects of the intervention were assumed beyond the childhood period, consistent with current evidence, which shows limited maintenance of intervention effects over time [31]. In other words, the intervention was assumed to reduce the number of children entering the progression model as overweight or obese, without modifying the transition probabilities themselves. The long-term number of adult obesity cases prevented was estimated by comparing adult obesity outcomes between the intervention and control scenarios. Transition probabilities were obtained from previous studies, considering gender [27, 29]. All model parameters are detailed in the Additional file 2 (Formula of calculation 1: The obesity progression model [23, 27, 29]).

The number of quality-adjusted life years (QALYs) gained per adult obesity case prevented was derived from a life-table approach, incorporating age- and sex-specific estimates of mortality risk, activity impairment, and life expectancy for obese vs. non-obese individuals [32]. We assumed that the health benefits from preventing adult obesity would begin at age 40 and persist through age 65. Based on literature-derived parameters [27, 29], we calculated the QALYs gained per obesity case prevented as the discounted difference in health utility over the expected lifetime between obese and non-obese adults, adjusted for survival probabilities. These included both quality-of-life (activity scale score) and life expectancy adjustments for those who die between 40 and 65 and those who survive. The average QALYs gained per case (Q) were

then multiplied by the estimated number of adult obesity cases prevented (N_{prevent}). A 5% annual discount rate was applied, consistent with pharmacoeconomic evaluation guidelines in China [33]. Formulas for calculating QALY saved were included in the Additional file 2 (Formula of calculation 2: QALY saved [27, 32, 33]).

Medical costs averted ($C_{\text{medical loss}}$) per obese adult were estimated according to previous studies [26, 27]. The parameter was derived from a Chinese study that estimated the economic costs of obesity [34]. The study used the data from the National Behavioral Risk Factors Surveillance Survey and the National Health Service Survey to estimate the total medical costs of five major non-communicable diseases related to overweight and obesity [34]. To calculate the medical cost per obese adult, we further used the data from the Population Census of the People's Republic of China to obtain the total number of obese adults [35]. This value was projected over 25 years (ages 40–65) and discounted at 5% annually to reflect lifetime medical cost savings per case prevented.

As for the cost of labor productivity loss ($C_{\text{productivity loss}}$), it included averted costs from lost productivity due to morbidity or mortality per obese adult prevented. The parameters in the formula were based on Brown's study [27]. Morbidity-related productivity loss was computed as the difference in annual absenteeism days between obese and non-obese adults, multiplied by sex-specific daily wages. Mortality-related loss was modeled as the difference in expected lifetime earnings between obese and non-obese adults from age 40 to 65, assuming premature mortality due to obesity. All estimates were adjusted for survival probability and discounted at 5%. Daily wage and annual wages in three cities (Beijing, Changzhi, and Urumchi) were calculated according to data from the National Bureau of Statistics of China [36]. The annual wage can be directly obtained from the statistics, while the daily wage is the annual income divided by the number of working days (365 days minus 104 statutory days off [37]). Formulas for averted medical costs and productivity loss costs were included in the Additional file 2 (Formula of calculation 3: Labor productivity loss [27, 32, 33, 36]; Formula of calculation 4: Medical costs averted [32, 33, 38–40]).

The incremental cost-utility ratio (ICUR) was calculated as the difference in total costs between the intervention and control groups divided by the difference in QALYs between the groups. The ICUR was calculated using the following formula:

$$ICUR = \frac{Net\ \text{intervention}\ cost}{N_{\text{Prevent}} \times Q}$$

where:

- $N_{Prevent} = Cases\ of\ adult\ obesity\ prevented$
- $Q = QALYs\ saved\ per\ case\ of\ obesity\ prevented\ through\ obesity\ prevention$

To assess the robustness of the long-term cost-utility analysis (CUA) results, we conducted probabilistic sensitivity analysis. We made the cost-effectiveness acceptability curve (CEAC) to show the relationship between the probability of the intervention being cost-effective and varying willingness-to-pay (WTP) thresholds per QALY gained. Probabilistic distributions were selected for each parameter based on its characteristics; the distribution of the parameters is detailed in the Additional file 3: Appendix Table 1 [41–43]. To account for parameter uncertainty, the range of variation for each parameter was defined as ± 20% of its base-case value [44]. We then performed 1000 independent Monte Carlo simulation trials using R 4.4.0 software.

For the cost–benefit analysis, we estimated the total economic benefit of the intervention as the sum of averted direct medical costs and indirect labor productivity loss, both per case of adult obesity prevented. The cost–benefit ratio (CBR) was calculated using the following formula:

$$CBR = \frac{\text{Net intervention cost}}{N_{\text{Prevent}} \times (C_{\text{medical loss}} + C_{\text{productivity loss}})}$$

where:

- $C_{medical\ loss} = medical\ costs\ averted\ per\ case\ of\ obesity\ prevented$
- $C_{productivity\ loss} = the\ cost\ of\ labor\ productivity\ loss\ averted\ per\ case\ of\ obesity\ prevented$

Projecting population-level impact and net economic benefits

To estimate the potential long-term population-level impact of scaling up the intervention nationally, we estimated net economic benefits through a five-step approach based on national demographic and educational statistics. First, we calculated the net effect of the intervention on childhood overweight and obesity prevalence by comparing changes between intervention and control groups. To adjust for potential attenuation in real-world implementation, we assumed a 10% reduction in intervention effectiveness due to implementation challenges during nationwide scale-up. Second, we projected the number of children affected using national enrollment statistics (Statistical Bulletin on Educational Development [45]) and sex-specific prevalence rates (the China Statistical Yearbook [46]). Third, we applied a two-stage obesity progression model to estimate the number of childhood cases likely to persist into adulthood.

Fourth, we quantified the lifetime economic burden per adult with obesity—including direct medical costs and productivity loss—to derive total potential cost savings. Finally, we compared the projected economic benefit to the national implementation cost of the intervention to calculate the overall net economic return. The net benefits (NB) from national rollout were calculated using the following formula:

$$NB = N_{\text{national prevented}} \times (C_{\text{medical loss}} + C_{\text{productivity loss}}) - \text{Net intervention cost}$$

Sensitivity analyses

We adjusted the costs of the intervention and the parameters for the model estimates for the potential scenario to simulate nationwide rollout. We assumed that each school would have a full set of teaching aids, increasing the number of sets from 3 to 12 and adjusting the cost accordingly (12×188.47 USD per set), and that all 12 schools would require a full set of physical activity supplies, with the cost adjusted to 12×78.41 USD per set. To generalize productivity costs beyond the study regions, daily wages were updated to the national average (315.68 CNY) and annual wages to 82,413 CNY. Following international discounting practices, we tested a lower discount rate of 3%, and to reflect potential reductions in real-world implementation, the intervention effect on obesity prevention was reduced by 10% relative to the base-case estimate. These assumptions reflect plausible variations in real-world implementation and long-term projections (Additional file 3: Appendix Table 2 [47]).

Results

Sample characteristics

A total of 24 schools were included in the study, with 12 schools assigned to the intervention group and 12 schools to the control group. The intervention group comprised 705 students, while the control group included 687 students. The mean age of participants was 9.6 years (SD 0.4) in both groups. The mean BMI was 18.5 (SD 3.7) in the intervention group and 18.8 (SD 3.7) in the control group, with corresponding BMI *Z*-scores of 0.7 (SD 1.4) and 0.8 (SD 1.4), respectively. The proportion of boys was 50.1% in the intervention group and 53.0% in the control group. Parents were the primary caregivers for the majority of children in both groups (95.1% and 96.2%), and 40.4% of parents in the intervention group and 42.4% in the control group had attained at least a high school education. No statistically significant differences were observed between groups at baseline, indicating comparability of the intervention and control groups.

Intervention costs

Monetary costs of the intervention were 8185.48 USD in total across all 12 intervention schools during the 1-year implementation period. Intervention costs were categorized into four domains according to target group and corresponding intervention measures: student-focused activities, family-focused activities, school-focused activities, and system/app-related activities. This corresponds to an average cost of 682.12 USD per school. The category "student-focused activities" accounted for the largest proportion (subtotal costs = 4238.93 USD, 51.79% of the total cost), including health education activities, physical activity supplies, and weight monitoring. The second major cost category was "family-focused activities" (subtotal costs = 2200.40 USD, 26.88% of the total cost), primarily covering the development and dissemination of various educational materials for parents. Another portion of the expenses was allocated to "school-focused activities" (subtotal costs = 992.25 USD, 12.12% of the total cost), which included preparing health-related bulletin boards, posters, and activity corners to create a supportive school environment. App-related operational costs—including system maintenance, routine updates, user data processing, and data storage—were calculated as a separate category, totaling 753.90 USD (9.21% of the total cost) (Table 1). The per-student intervention material cost was 11.08, 12.58, and 11.39 USD in Beijing, Changzhi, and Urumqi, respectively, with an overall average of 11.61 USD across all sites.

As shown in Table 2, the total labor cost was 5584.26 USD. Among staff categories, class teacher accounted for the largest share (subtotal costs = 2205.72 USD, 39.50% of the total cost), followed by principals (subtotal costs = 1365.16 USD, 24.45% of the total cost), healthcare or health education teachers (subtotal costs = 1084.64 USD, 19.42% of the total cost), and PE teachers (subtotal costs = 928.74 USD, 16.63% of the total cost). By region, the labor cost of the three intervention areas of Beijing, Changzhi, and Urumqi was respectively 2276.38, 1316.56, and 1991.32 USD. The corresponding per-student labor costs were 8.76, 6.58, and 8.13 USD, with an overall average of 7.92 USD.

The total cost of implementing this intervention across all 12 schools over the 1 year was 13,769.74 USD, with 5157.14, 3831.92, and 4780.73 USD incurred in Beijing, Changzhi, and Urumqi, respectively. Based on the total number of students who participated in the intervention ($n = 705$), the cost per student was estimated at 19.53 USD, with site-specific costs of 19.84, 19.16, and 19.52 USD, respectively.

Table 1 Monetary costs of the Diet, ExerCIse, and CarDiovascular hEalth-Children intervention

Major categories	Items	Unit cost (USD)	Beijing (*N* = 260)		Changzhi (*N* = 200)		Urumqi (*N* = 245)		Total quantity	Subtotal costs (USD)
			Quantity	Cost (USD)	Quantity	Cost (USD)	Quantity	Cost (USD)		
Supportive school environment	The book "Eat properly, healthy weight"	2.84	32	90.88	32	90.88	32	90.88	96	272.85
	APP User Manual for teachers	0.6	8	4.80	8	4.80	8	4.80	24	14.47
	The manual "Health Education" for class teachers	1.09	8	8.72	8	8.72	8	8.72	24	26.05
	The manual "An Operation Manual for School Team Members Involved in the Multi-component Obesity Intervention among Primary School Students"	1.16	32	37.12	32	37.12	32	37.12	96	111.46
	The book "Exercise Health Guidance for children and Adolescents"	3.77	8	30.16	8	30.16	8	30.16	24	90.47
	Weighing scales	4.04	8	32.32	8	32.32	8	32.32	24	96.98
	Physical examination feedback sheets for school staff	0.02	400	8.00	400	8.00	400	8.00	1200	18.09
	Posters	30.16	4	120.64	4	120.64	4	120.64	12	361.87
	Subtotal			332.64		332.64		332.64		992.25
Supportive family environment	APP User Manual for Parents	0.3	260	78.00	200	60.00	245	73.50	705	212.6
	Feedback materials	0.3	780	234.00	600	180.00	735	220.50	2115	637.8
	Materials for winter vacation	0.17	260	44.20	200	34.00	245	41.65	705	116.93
	Physical examination feedback sheets for parents	0.15	2080	312.00	1600	240.00	1960	294.00	5640	850.4
	Weekly home exercise record sheet for parents	0.54	260	140.40	200	108.00	245	132.30	705	382.68
	Subtotal			808.60		622.00		761.95		2200.4
Student-focused activities	Materials for "Small hand in big hand" homework	0.14	260	36.40	200	28.00	245	34.30	705	95.67
	The book "Eat properly, healthy weight"	2.84	260	738.40	200	568.00	245	695.80	705	2003.75
	Teaching aids	188.47	1	188.47	1	188.47	1	188.47	3	565.42
	Gifts	71.54	4	286.16	4	286.16	4	286.16	12	858.54
	Basketball, football, skipping rope, shuttlecock, etc	78.41	2.67	209.35	2.67	209.35	2.67	209.35	8	627.24
	Weekly weight monitoring sheets	0.02	32	0.64	32	0.64	32	0.64	96	1.45
	Weekly weight feedback sheets	0.45	64	28.80	64	28.80	64	28.80	192	86.85
	Subtotal			1488.22		1309.42		1443.52		4238.93
App	System maintenance, routine updates, user data processing, and data storage			251.30		251.30		251.30		753.9
Total costs				2880.76		2515.36		2789.41		8185.48
Cost per student				11.08		12.58		11.39		11.61

Table 2 Labor costs of the Diet, ExerCIse, and CarDiovascular hEalth-Children intervention

Major categories	Beijing (*N* = 260)		Changzhi (*N* = 200)		Urumqi (*N* = 245)		All
	Unit cost[a] (USD)	Quantity[b]	Unit cost[a] (USD)	Quantity[b]	Unit cost[a] (USD)	Quantity[b]	Subtotal costs (USD)
Principals	20.94	32	10.47	28	12.56	32	1365.16
Class teachers	15.71	64	8.38	58	11.52	62	2205.72
Healthcare or health education teachers	12.56	24	7.33	44	10.47	44	1084.64
PE teachers	13.61	22	5.24	41	9.42	44	928.74
Total cost		2276.38		1316.56		1991.32	5584.26
Cost per student		8.76		6.58		8.13	7.92

[a] USD/person-hour, the hourly salary, calculated by dividing the annual salary of teachers at different levels in the district by 9 months (3 months for winter and summer vacation) and then by 8 h (8 working hours per day on weekdays)

[b] Labor hours, obtained from the program process records

Incremental cost-effectiveness ratios

After the 9-month intervention, the intervention group had significant improvements in obesity-related outcomes compared to the control group, including significant reductions in BMI (adjusted mean difference − 0.46 [95% CI − 0.67 to − 0.25]), BMI *Z*-score (adjusted mean difference − 0.17 [95% CI − 0.25 to − 0.09]), BF% (adjusted mean difference − 1.05 [95% CI − 1.83 to − 0.28]), and WC (adjusted mean difference − 1.63 [95% CI − 2.82 to − 0.43]) [23]. Using these observed mean differences and the calculated per-student intervention cost, the incremental cost-effectiveness ratios were estimated as follows: 42.46 USD (95% CI 29.15 to 78.12) per unit reduction in BMI, 11.49 USD (95% CI 7.81 to 21.70) per 0.1 unit reduction in BMI *Z*-score, 18.60 USD (95% CI 10.67 to 69.75) per 1% reduction in body fat percentage, and 11.98 USD (95% CI 6.93 to 45.42) per 1 cm reduction in waist circumference (Table 3).

Cost-utility and cost–benefit analyses

The results of cost-utility and cost–benefit analyses are shown in Table 4. According to the Chinese reference standard for obesity, the intervention was estimated to have prevented 26 cases of childhood overweight/obesity among the intervention group during the 9-month study period. This corresponds to an estimated cost of 529.61 USD per overweight/obesity case averted. Applying the two-stage obesity progression model to these averted cases, we estimated that 11.84 cases of adult obesity could be prevented in the long term. The analyses estimated 2.97 QALYs gained. This resulted in an ICUR of 4644.42 USD per QALY gained. Economic returns were generated through reductions in healthcare expenditures and productivity losses associated with obesity. Specifically, averted medical costs associated with obesity were estimated at 4004.46 USD, while productivity losses avoided were 11,728.11 USD, based on the number of adult obesity cases prevented and the respective average costs per case. The cost-effectiveness acceptability curve indicated that the intervention would be cost-effective (probability was 95%) at a willingness-to-pay threshold of 5200 USD/QALY (Fig. 1). Combining these benefits, the CBR was calculated as 0.84, indicating that for every 1 USD invested, 0.84 USD of direct and indirect

Table 3 Incremental cost-effectiveness ratios (ICERs) for anthropometric significant outcomes of the Diet, ExerCIse, and CarDiovascular hEalth-Children intervention

Outcomes	Intervention vs control adjusted mean difference (95% CI)[a]	ICER (USD per unit change; 95% CI)	Interpretation
BMI (kg/m^2)	− 0.46 (− 0.67 to − 0.25)	42.46 (29.15 to 78.12)	Cost to prevent 1 unit increase in BMI
BMI *Z*-score	− 0.17 (− 0.25 to − 0.09)	11.49 (7.81 to 21.70)	Cost to prevent 0.1 unit increase in BMI *Z*-score
Body fat (%)	− 1.05 (− 1.83 to − 0.28)	18.60 (10.67 to 69.75)	Cost to prevent 1 percentage point increase in body fat
Waist circumference (cm)	− 1.63 (− 2.82 to − 0.43)	11.98 (6.93 to 45.42)	Cost to prevent 1 cm increase in waist circumference

BMI body mass index

[a] Adjusted for baseline value and covariates

Table 4 The results of the cost-utility and cost–benefit analyses for the Diet, ExerCIse, and CarDiovascular hEalth-Children intervention

Indicators	Chinese obesity reference	WHO obesity reference
Overweight/obesity cases prevented in the intervention	26	15
Cost of avoiding one overweight/obesity case	529.61	917.98
Cases of adult obesity prevented	11.84	6.83
Net intervention cost (USD)	13,769.74	
QALYs gained	2.97	1.48
Medical costs averted (USD)	4004.46	2310.00
Productivity loss averted (USD)	11,728.11	6765.46
ICUR (USD)	4644.42	9288.83
CBR	0.84	1.45

QALY quality-adjusted life years, *ICUR* incremental cost-utility ratio, *CBR* cost–benefit ratio

costs were offset. In a national-scale scenario analysis, we projected that implementing the intervention across all primary schools in China could prevent approximately 3.61 million cases of childhood overweight/obesity. Using the same progression model, this would translate into 1.70 million adult obesity cases averted and an estimated 419,040 QALYs gained (0.249 per female case and 0.238 per male case). The total net economic benefit was estimated at 1.86 billion USD, highlighting the substantial potential for health and economic gains through national rollout (Additional file 4: Calculating procedure).

Sensitivity analyses

The sensitivity analysis indicated that this intervention measure remained cost-effective in all possible scenarios, including simulations of nationwide promotion, such as

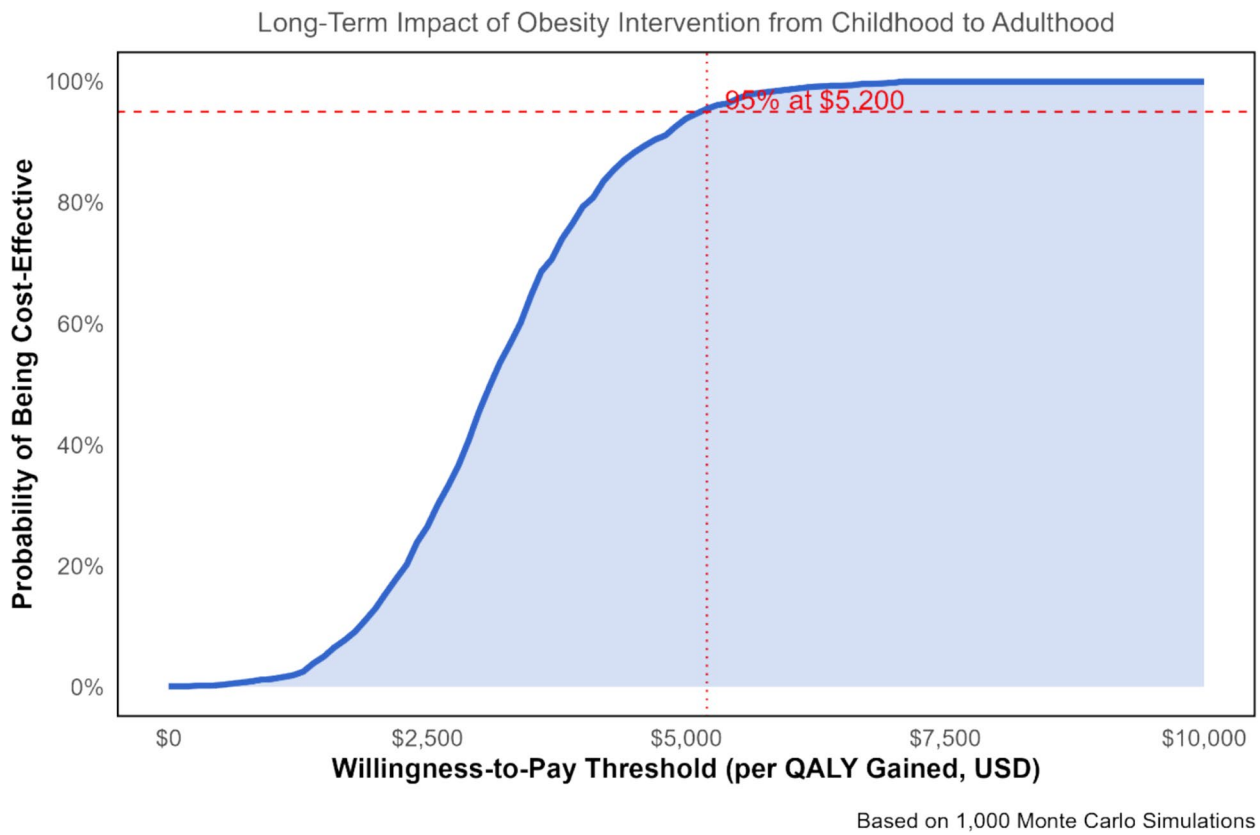

Fig. 1 Cost-effectiveness acceptability curve

increasing the number of teaching aids, adjusting personnel costs to the national average level, or adopting a lower discount rate of 3%. Although the hypothetical scenario in which the intervention effect weakens leads to a decrease in cost-effectiveness and cost-utility, this does not necessarily mean that the value of large-scale implementation will be lost. During the process of large-scale promotion, process optimization and resource sharing are likely to reduce the per capita intervention cost, which may potentially offset the weakening of the effect and maintain the economic feasibility of the intervention measure (Additional file 3: Appendix Table 3–6).

Discussion

The DECIDE-Children intervention demonstrated significant improvements in obesity-related anthropometric outcomes among school-aged children [23]. To assess the economic efficiency of this multi-component program—which incorporated strong parental engagement and mobile health support—we conducted a comprehensive health economic evaluation. We found that the intervention cost was 19.53 USD per student. The ICERs were 42.46 USD per unit reduction in BMI and 11.49 USD per 0.1 unit reduction in BMI *Z*-score, respectively. Cost-utility and cost–benefit analyses further indicated that the intervention yielded 2.97 quality-adjusted life years (QALYs), with an ICUR of 4644.42 USD and a CBR of 0.84. If scaled nationally, the intervention was estimated to gain 419,040 QALYs and yield a net economic benefit of 1.86 billion USD, highlighting its potential to reduce long-term healthcare expenditure. These findings remained robust in sensitivity analyses.

To directly evaluate the intervention within the Chinese context, we conducted a probabilistic sensitivity analysis to estimate the probability of the intervention being cost-effective across a range of WTP thresholds. The resulting cost-effectiveness acceptability curve shows that the probability of cost-effectiveness reaches 95% at a WTP threshold of US$5200 per QALY gained. This threshold is well below China's commonly used threshold (typically 1–3 times the GDP per capita [48], which was approximately US$10,085 in 2018). Therefore, the DECIDE-Children intervention has a high probability of being cost-effective and represents a financially viable strategy for scalability within China's primary school system.

Two main factors contributed to the low cost of our intervention. First, the intervention was built into the existing school health and PE classes, making it easier to carry out without adding extra pressure on teachers or students. Carter et al. [49] emphasized that interventions may become overburdened when extensive coordination and overhead costs are incurred, particularly with poor utilization of available capacity. Thus, embedding obesity prevention within standard school frameworks is both cost-effective and scalable. Second, we used a mobile application to facilitate communication between schools and families, significantly reducing the need for labor-intensive, in-person parent workshops and associated costs (e.g., transportation subsidies). This mHealth-supported model provided compelling evidence for the cost-effectiveness of digital tools in obesity intervention delivery.

It is important to note that our estimates of economic benefit were conservative and likely underestimated the full value of the intervention. First, lifestyle improvements in diet and physical activity have independent effects on chronic diseases [50], meaning that health gains may extend beyond weight reduction [51]. Even a 0.1 unit reduction in the BMI *Z*-score has been shown to improve cardiometabolic risk in children who were overweight or obese. Second, addressing childhood obesity also enhances psychosocial outcomes—including quality of life, social relationships, and academic performance—all of which offer additional societal benefits [52]. Third, since many intervention components were delivered through caregivers, the program likely generated spillover effects by improving the health literacy and behaviors of family members [52]. Fourth, the analysis excluded potential savings in pediatric healthcare costs, thus underestimating the averted medical costs [53].

This study has several strengths. First, we employed a multi-center cluster RCT conducted across three provinces in China with different economic levels, which allows the cost-effectiveness results to have a certain degree of national representativeness. Second, the costs associated with the intervention in this study were collected prospectively through direct recording by investigators and school personnel, with structured forms and regular checks implemented to minimize recall bias. Third, our study employed multiple anthropometric indicators (BMI, BMI *Z*-score, waist circumference, body fat percentage), whereas most previous economic evaluations of school-based interventions focused primarily on behavioral outcomes or reported only one or two anthropometric measures, limiting comparability and real-world relevance. In a systematic review of economic evaluations of school-based lifestyle interventions, only 8 of the 23 studies used weight-related measures [54]. BMI and BMI *Z*-score are considered to be more valid outcome measures than behavior change measures in obesity intervention studies and more suitable for health economic models to infer and evaluate the long-term cost-effectiveness of interventions [54]. At the same time, to enhance generalizability and comparability across different contexts and health conditions, we also included

QALYs as a standardized outcome measure that captures overall quality-adjusted survival. Fourth, in addition to considering immediate cost-effectiveness, we adopted a lifetime perspective (up to age 65) to evaluate the intervention's broader impact on adult disease burden and workforce productivity.

Despite these strengths, several limitations should be noted. First, our economic evaluation used clinical outcomes (e.g., BMI *Z*-score) rather than a generic measure like the quality-adjusted life-year (QALYs), which may limit comparability with interventions targeting other health conditions. Second, we did not account for caregiver time and productivity losses, such as time spent attending educational sessions. However, these costs were likely minimal and partially offset by the flexibility of the App-based delivery, which allowed parents to participate in their free time. Similarly, we also did not include the opportunity cost of children's time. This was because the intervention activities (e.g., health education, physical activity breaks) were fully integrated into the standard school curriculum and schedule, and thus did not displace additional academic or leisure time from the children's perspective. Nevertheless, we acknowledge that from a strict societal perspective, this represents a limitation of our cost analysis. Third, although the intervention demonstrated significant short-term effectiveness, the long-term sustainability of these effects—such as continued improvements or maintenance in weight status—remains uncertain and warrants further longitudinal follow-up. Fourth, the trial was implemented in selected provinces with different overall economic levels, which allows the study to have a certain degree of national representativeness. However, the findings may not be generalizable to all regions of China due to potential variations in school infrastructure, local policies, and parental engagement. Finally, the study was originally powered based on primary effectiveness outcomes (BMI and BMI *Z*-score), rather than economic endpoints. Therefore, it may have limited power to detect subtle differences in cost-effectiveness or cost-utility outcomes, which should be considered when interpreting the results.

Conclusions

The DECIDE-Children intervention demonstrated strong effectiveness in preventing childhood obesity, with higher cost-effectiveness compared to previously reported school-based programs. The intervention yielded substantial net economic benefits by lowering projected medical expenditures and productivity losses associated with obesity-related non-communicable diseases. Its integration of school, family, and digital components offers a scalable and sustainable approach to tackling childhood obesity. Given its effectiveness, cost-efficiency, and feasibility, this multi-component intervention holds promise for national implementation in China and may serve as a valuable model for other developing countries facing similar public health challenges.

Abbreviations

BF%	Body fat percentage
BMI	Body mass index
BMI *Z*-score	Body mass index-for-age *Z*-score
CBR	Cost–benefit ratio
CEAC	Cost-effectiveness acceptability curve
CHEERS	Consolidated Health Economic Evaluation Reporting Standards
CUA	Cost-utility analysis
DECIDE-Children	Diet, Exercise, and Cardiovascular Health–Children
ICER	Incremental Cost-Effectiveness Ratio
ICUR	Incremental cost-utility ratio
NB	Net benefit
QALYs	Quality-adjusted life year
RCT	Randomized controlled trial
WC	Waist circumference
WHO	World Health Organization
WTP	Willingness-to-pay

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12916-025-04451-x.

Additional File 1. CHEERS 2022 Checklist. A completed CHEERS 2022 checklist documenting the compliance of this study with the reporting standards for economic evaluations.

Additional File 2. Key Calculation Formulas. This file contains the mathematical formulas used in the analysis: Formula 1 – The obesity progression model; Formula 2 – Calculation of QALYs saved; Formula 3 – Estimation of labor productivity loss; Formula 4 – Calculation of medical costs averted.

Additional File 3. Assumptions, Parameter Distributions, and Scenario Analyses. This file includes the supplementary tables summarizing assumptions, parameter distributions, and scenario analyses: Appendix Table 1 – Parameter distribution explanation; Appendix Table 2 – Scenario assumptions and cost adjustments in sensitivity analyses; Appendix Table 3 – Results of cost-utility and cost–benefit analyses (Scenarios 1–3); Appendix Table 4 – Results with adjusted discount rate (Scenario 4: 3% vs. 5% base case); Appendix Table 5 – Results with reduced intervention effectiveness (Scenario 5: − 10% reduction in obesity prevention rate); Appendix Table 6 – Combined results for Scenarios 4 and 5.

Additional File 4. Calculation Procedure for Long-term Impact. This file provides a detailed description of the stepwise procedure for estimating the long-term, population-level impact of scaling up the intervention nationally, including projections of obesity cases averted and associated health-economic outcomes.

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Authors' contributions

S.-Y.Y. conducted the analysis and wrote the original draft, S.Z. did the investigation and data collection and curation. X.-RG., J.-L.L., H.W. and H.F. revised the manuscript. H.-J.W. and H.F. supervised the study. All authors read and approved the final manuscript.

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Data availability
Data from this study would be available by contacting the corresponding authors and could be provided with the consent of the Ethics Committee.

Declarations

Ethics approval and consent to participate
The study was approved by the Institutional Review Board of Peking University (IRB00001052-18021) and registered at www.clinicaltrials.gov (NCT03665857) (accessed on 1 March 2022). Written informed consent was obtained from all subjects participating in the study. The study was conducted according to the Declaration of Helsinki.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

Author details
[1]Department of Maternal and Child Health, School of Public Health, National Health Commission Key Laboratory of Reproductive Health, Peking University Health Science Center-Weifang Joint Research Center for Maternal and Child Health, Peking University, Beijing, China. [2]China Center for Health Development Studies, National Health Commission Key Laboratory of Reproductive Health, School of Public Health, Peking University, Beijing, China.

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